EE 354/CE 361/MATH 310 -

Introduction to Probability

and Statistics

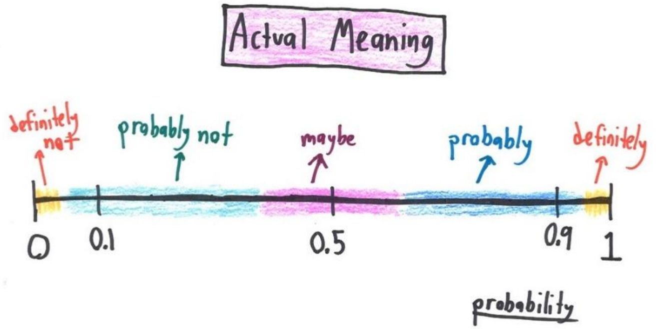
Fall 2025

EE 354/CE 361/MATH 310 L2 section (Fall 2025)

Introduction to Probability and Statistics -

Aamir Hasan

Week 5 - Lecture No 09 – 20th September 2025



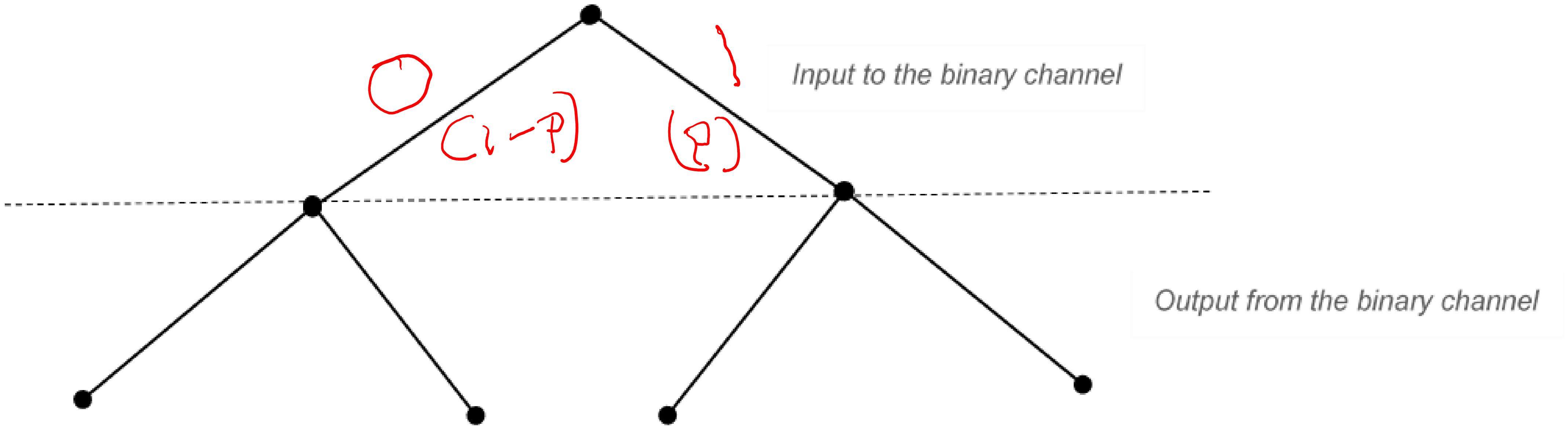
Announcements!

1. Lectures up to No 08 posted
2. An added lecture will be posted in a asynchronous mode in lieu of September 08, 2025 class
3. Quiz No 03 – September 24, 2025 – on Counting
4. TA's

Agenda for today

1. Quiz No 02
2. Unit 3 - Counting

Quiz No 2


0
1
(1-p)
(p)
Input to the binary channel
Output from the binary channel

Meet Your TAs

Saif Nazir

About Me

I am a Computer Engineering Senior. I am interested in Robotics and AI. Apart from Academics, I play Tabla and Sitar and I am also part of the volleyball team at Habib. You can always find me either in the Music room or at the courts.

Ehsaas Hours

- Tue 12:00 - 1:30
- Thu 12:00 - 1:30

Contact Me

sno8448@st.habib.edu.pk or “Saif Nazir” on Teams

Meet Your TAs

Rayyan Shah

About Me

I am a Computer Science Junior. I am interested in bioinformatics, AI, and game development. Outside of academics, I also enjoy cooking and baking, and I always have some new recipe to recommend. You can usually find me in the library or tapal.

Ehsaas Hours

- Mon 11:30 - 12:30
- Tue 2:30 - 3:30
- Fri 3:30 - 4:30

Contact Me

rs08770@st.habib.edu.pk

Or “Rayyan Ali Shah” on Teams

Meet Your TAs

Hiba Aiman Ali

About Me

I am a Computer Engineering Junior with a minor in South Asian Music. My interests include bioinformatics, IoT, music and sustainability. I also sing as a part of the Habib University Orchestra. When I’m not in class, I’m most likely in the music room, singing or *trying to* learn an instrument.

Ehsaas Hours

- Tue 4:00 - 5:00
- Fri 8:30 - 10:30

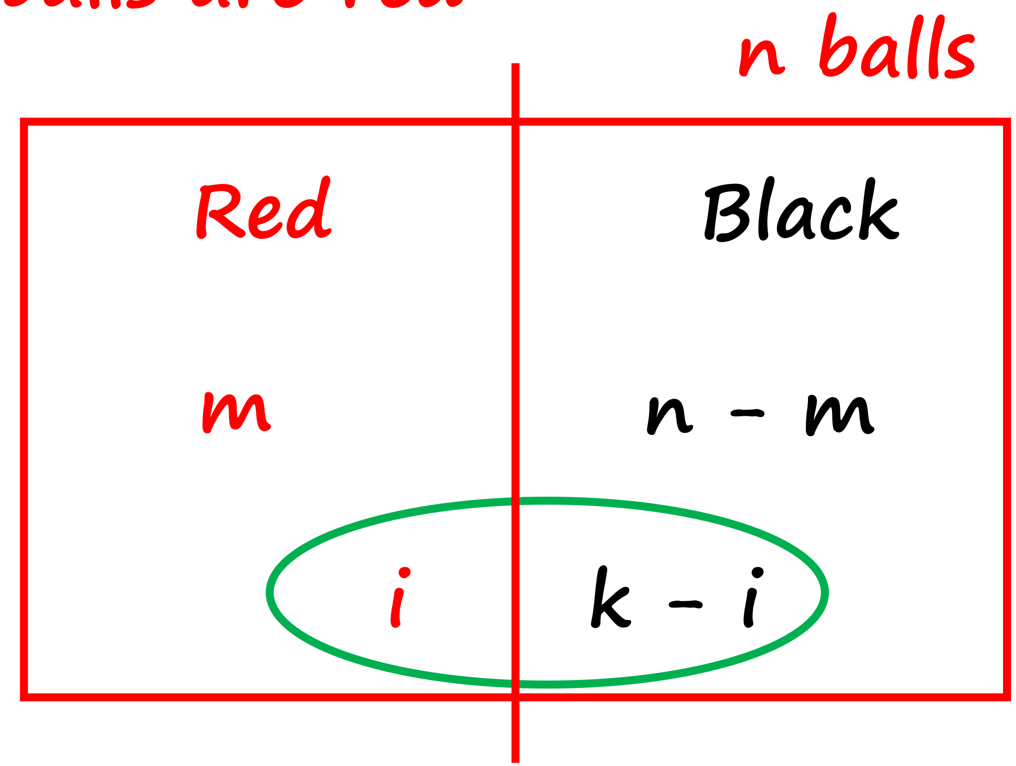
Contact Me

ha09269@st.habib.edu.pk on Teams or Outlook

Counting Example

- We have a total of n balls, out of which m are red balls.
  The remaining balls $(n-m)$ are black balls
- We have to pick k balls such that i balls are red

Find the Probability,
P (i red balls) $= P_r$



Counting Example

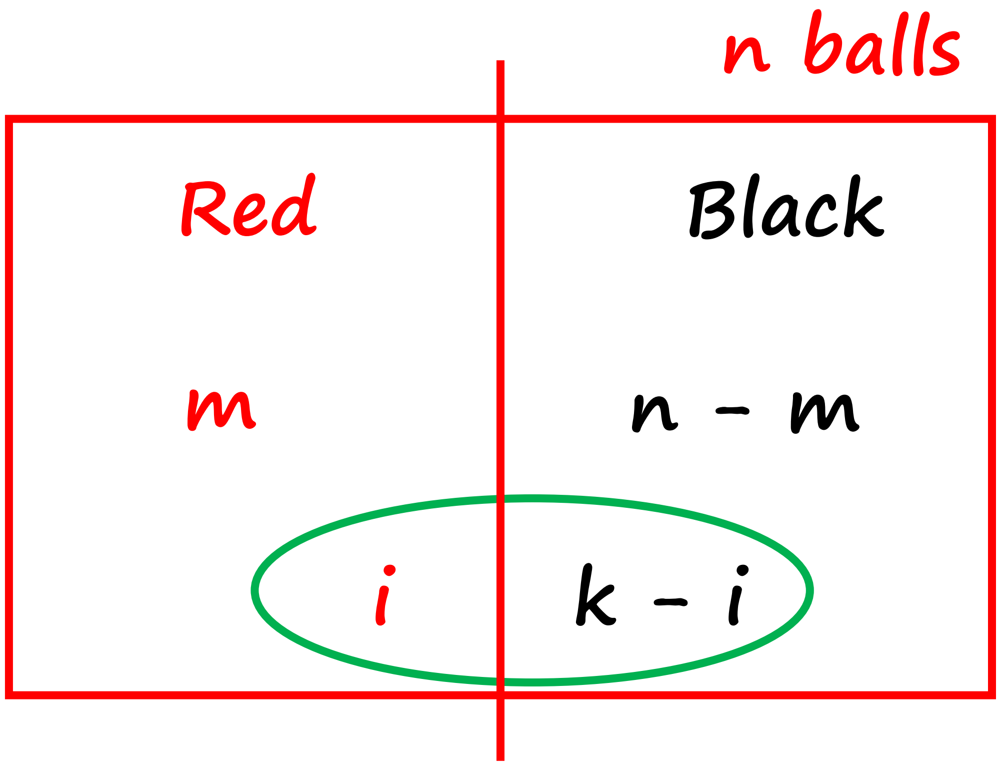
$P(i \text{ red balls}) = P_r$

$|\Omega| = \binom{n}{k}$

➡ Let C = # of ways to get i red balls in k balls

$= a * b$

- a = # of ways to get i out of m red balls
- b = # of ways to get $(k-i)$, out of $(n-m)$, black balls



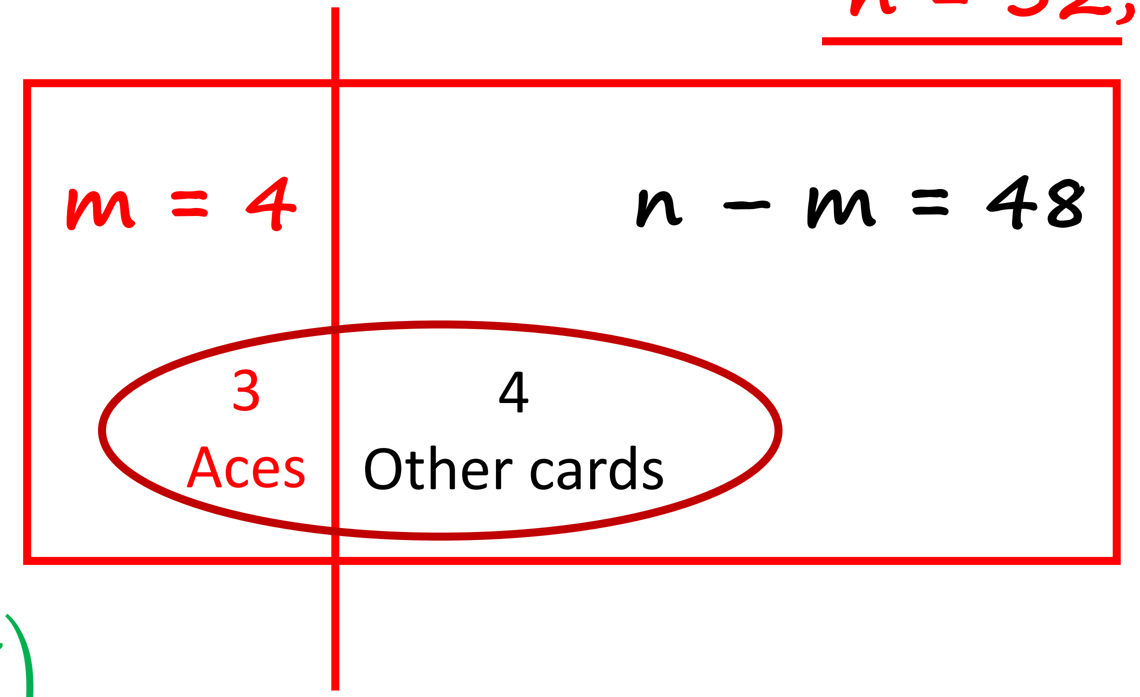
$$P_r = \frac{C}{|\Omega|} = \frac{\binom{m}{i}\binom{n-m}{k-i}}{\binom{n}{k}}$$

Counting Example

In a deck of 52 cards, you select 7 cards.
What is the probability that out of 7 cards three are aces, P (3 aces)?

$$P_r = \frac{C}{|\Omega|} = \frac{\binom{m}{i}\binom{n-m}{k-i}}{\binom{n}{k}}$$

$n = 52$, $k = 7$, $i = 3$



$$|\Omega| = \binom{n}{k}$$

$$C = a * b$$

$$P\,(3 \text{ aces}) = \frac{\binom{4}{3}\binom{48}{4}}{\binom{52}{7}}$$

The multinomial Probabilities

- Balls of different colors: i = 1, . . . , r
- Probability of picking a ball color i is p_i
- Draw n balls, independently
- Given nonnegative numbers n_i, with $n_1 + \ldots + n_r = n$
- Find: $\mathbf{P}(n_1 \ balls \ of \ color \ 1, n_2 \ balls \ of \ colors \ 2, \ldots, n_r balls \ of \ color \ r \ )$
- Special case r = 2; colors: "heads" , "tails"

p $1-p$

$$\binom{n}{k} p^k (1-p)^{n-k}$$

$n_1 \longleftrightarrow$ k heads $p_1 = p$

$n_2 \longleftrightarrow$ n-k heads $p_2 = 1 - p$

$$\frac{n!}{n_1!\, n_2!} p_1^{n_1} p_2^{n_2}$$

The multinomial Probabilities

$\boldsymbol{n_1 + \ldots + n_r = n}$

3

2

1

1 2 3 4 5 6 7

$r = 3$

$n = 7$

1 1 3 1 2 2 1

Type: (4, 2, 1)

Prob $= p_1 \,.\, p_1 \,.\, p_3 \,.\, p_1 \,.\, p_2 \,.\, p_2 \,.\, p_1 = p_1^4 \,.\, p_2^2 \,.\, p_3$

$\mathbf{P}$(particular sequence of "type"$(n_1, n_2, \ldots, n_r)$) = $p_1^{n_1} p_2^{n_2} p_3^{n_3} \ldots p_r^{n_r}$

sequence of "type"$(n_1, n_2, \ldots, n_r)$ $\longleftrightarrow$ Partition of {1, . . . , n}} into subsets of sizes $n_1, n_2, \ldots, n_r$

$$\mathbf{P}(\text{ get type } (n_1, n_2, \ldots, n_r)) = \frac{n!}{n_1!\, n_2! \;\; \ldots n_r!} p_1^{n_1} p_2^{n_2} p_3^{n_3} \ldots p_r^{n_r}$$

Practise Problem 1 –

- An experiment consists of rolling a 20 sided die three times. The number on top of each die is recorded. The numbers are written down in the order in which they are observed. How many possible ordered triples of numbers can result from the experiment? (Note the triple (17, 10, 3) is not the same result as the triple (3, 10, 17)).

Answer:-

There are 20 ways each throw can come up and the order is important so the answer is $20 \cdot 20 \cdot 20 = 20^3 = 8000$.

Practise Problem 2

- When you buy a Powerball ticket, you select 5 different white numbers from among the numbers 1 through 59 (order of selection does not matter), and one red number from among the numbers 1 through 35. How many different Powerball tickets can you buy?

Answer:-

you need to select 5 distinct white numbers, so you can do this $C(59, 5) = 5,006,386$ ways. Then you can pick the red number $C(35, 1) = 35$ ways so the total number of tickets is:-

$C(59, 5) \cdot C(35, 1) = 5,006,386 \cdot 35 = 175,223,510$.

Practise Problem 3

- If I flip a coin 20 times, I get a sequence of Heads (H) and tails (T). How many different sequences have at least three heads?

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Solution: The total number of sequences that are possible here is 2^{20}. We want number of sequences that have at least 3 heads, so we can find out the number of sequences that have no heads, 1 heads, and 2 heads and subtract it from total number of sequences.

$$\text{Number of sequences with no heads} = 1$$

$$\text{Number of sequences with 1 heads} = \binom{20}{1}$$

$$\text{Number of sequences with 2 heads} = \binom{20}{2}$$

Therefore,

$$\text{Number of sequences with at least 3 heads} = 2^{20} - 1 - \binom{20}{1} - \binom{20}{2}$$

We can also approach it by finding the number of sequence with 3 heads, 4 heads, all the way to 20 heads and summing them up, i.e.

$$\text{Number of sequences with at least 3 heads} = \sum_{k=3}^{20} \binom{20}{k}$$

Practise Problem 4

- From 5 seniors, 4 juniors, 2 sophomores and 9 freshmen, in how many ways can the group of five be chosen if there must be at least one member from each class?

Answer

There are 5 ways to select a senior, 4 ways to select a junior, 2 ways to select a sophomore and 9 ways to select a freshman. This gives $5 \cdot 4 \cdot 2 \cdot 9 = 360$ ways to select a subset with 4 elements containing one member of each class. When you have done this there are $20 - 4 = 16$ members left and you may choose any one of these to round out the group. Hence the answer is $360 . 16/2 = 2,880$. You must divide by 2 because each set of 5 elements selected by this procedure occurs twice

Practise Problem 4 –

Some more explanation

❖ From 5 seniors, 4 juniors, 2 sophomores and 9 freshmen, in how many ways can the group of five be chosen if there must be at least one member from each class?

Answer

Here is another approach.

Because there are 5 members in the subset and 4 classes, exactly one class occurs twice.

– If there are 2 seniors, these can be selected in $C(5, 2)$ ways and the set filled out with 1 junior, 1 sophomore and 1 freshman, hence in $C(5, 2) \cdot 4 \cdot 2 \cdot 9 = \underline{720 \text{ ways.}}$

– If there are 2 juniors, these can be selected in $C(4, 2)$ ways and the set filled out with 1 senior, 1 sophomore and 1 freshman, hence in $5 \cdot C(4, 2) \cdot 2 \cdot 9 = \underline{540 \text{ ways.}}$

– If 2 sophomores, $5 \cdot 4 \cdot C(2, 2) \cdot 9 = \underline{180.}$

– If 2 freshmen, $5 \cdot 4 \cdot 2 \cdot C(9, 2) = \underline{1{,}440.}$

Hence the answer is $720 + 540 + 180 + 1{,}440 = 2{,}880$.

Practice Problem No 5: Recall that a standard deck of cards has 52 cards. The cards can be classified according to suits or denominations. There are 4 suits, hearts, diamonds, spades and clubs. There are 13 cards in each suit. There are 13 denominations, Aces, Kings, Queens, ... ,Twos, with 4 cards in each denomination. A poker hand consists of a sample of size 5 drawn from the deck. Poker problems are often like urn problems, with a hitch or two.

- Part a: How many poker hands consist of 2 Aces and 3 Kings?
- Part b: If a poker hand has three cards from one denomination and two from another it is called a full house. Find the Probability of getting a Full House, P_{FH}.
- Part c: A royal flush is a hand consisting of an Ace, King, Queen, Jack and Ten, where all cards are from the same suit. Find the Probability of getting a royal flush, P_{RF} .

Practice Problem No 5: Recall that a standard deck of cards has 52 cards. The cards can be classified according to suits or denominations. There are 4 suits, hearts, diamonds, spades and clubs. There are 13 cards in each suit. There are 13 denominations, Aces, Kings, Queens, ... ,Twos, with 4 cards in each denomination. A poker hand consists of a sample of size 5 drawn from the deck. Poker problems are often like urn problems, with a hitch or two.

- **Part a: How many poker hands consist of 2 Aces and 3 Kings?**

Ace's $\binom{4}{2}$ & 3 King $\binom{4}{3}$

$= 6 \cdot 4 = 24$ ways.

- **Part b: If a poker hand has three cards from one denomination and two from another it is called a full house. Find the Probability of getting a Full House, P_{FH}.**

There are 13 ways to Pick the first denomination and One can choose $\binom{4}{3}$ ways of Select 3 Cards of that denomination. There are 12 ways of Select the 2nd denomination. and $\binom{4}{2}$ of Select the 2 cards out of 4.

$$= 13 \cdot \binom{4}{3} \times 12 \times \binom{4}{2} = 3744$$

$$P_{FH} = \frac{3744}{\binom{52}{5}} = \frac{3744 \times 5!}{48 \times 49 \times 50 \times 51 \times 52}$$

$$= 0.00144$$

- Part c: A royal flush is a hand consisting of an Ace, King, Queen, Jack and Ten, where all cards are from the same suit. Find the Probability of getting a royal flush, P_{RF} .

There are 4 Royal flush's in the deck and within Each 1 way of Select the Cards. 3

$$\frac{4}{\binom{52}{5}} = \frac{4 \times 120}{48 \times 49 \times 50 \times 51 \times 52}$$

$$= 1.53 \times 10^{-6}$$